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个人简介

研究聚焦于大语言模型的后训练与对齐——以强化学习 (Off-Policy 稳定性、探索机制) 与在线蒸馏 (On-Policy Distillation) 为核心训练范式, 并在大模型的安全对齐与可解释性方向开展研究. 博士期间曾从事脑启发神经网络与认知科学研究, 为当前工作提供了跨学科视角. 以第一作者在 *PNAS*, *Cell Patterns* 等期刊以及 *ICLR*, *NeurIPS*, *CVPR*, *ICCV*, *AAAI* 等会议发表论文十余篇 (含 CCF-A 会议**七**篇). 总引用量 900+, H-Index 17.

教育经历

2021 – 2026 **Ph.D.**, 模式识别与智能系统 中国科学院自动化研究所
Bio-Inspired AI / LLM Alignment, Safety & Interpretability

2017 – 2021 **B.Eng.**, 通信工程 中山大学电子与信息工程学院
GPA: 4.05 / 5.0, **Rank**: 1 / 85 (综合)

实习经历

2025.10 – 至今 **顶尖实习生 (RedStar Intern)**, 小红书 **Hi Lab**, 北京
LLM 强化学习算法: 稳定 Off-policy RL 训练范式
On-Policy Self-Distillation: 无需外部 teacher 的在线自蒸馏训练范式
多模态后训练: 基于 GeoGuessr 视觉地理推理任务的后训练闭环

代表性论文

Large Language Models: Alignment, Safety & Post-Training

1. **Shen, Guobin**, Zhao, Chenxiao, Cheng, Xiang, Huang, Lei, and Yu, Xing. "VESPO: Variational Sequence-Level Soft Policy Optimization for Stable Off-Policy LLM Training." *arXiv preprint arXiv:2602.10693*, 2026. RL Off-Policy HF Weekly #3

2. **Shen, Guobin**, Zhao, Dongcheng, Tong, Haibo, Li, Jindong, Zhao, Feifei, and Zeng, Yi. "Safety Instincts: LLMs Learn to Trust Their Internal Compass for Self-Defense." *Proceedings of the 14th International Conference on Learning Representations*, 2026. AI Safety Instincts Rewards ICLR

3. **Shen, Guobin**, Zhao, Dongcheng, Dong, Yiting, He, Xiang, and Zeng, Yi. "Jailbreak Antidote: Runtime Safety-Utility Balance via Sparse Representation Adjustment in Large Language Models." *Proceedings of the 13th International Conference on Learning Representations*, 2025. LLM Safety Interpretability ICLR

代表性论文 (continued)

4. **Shen, Guobin**, Zhao, Dongcheng, He, Xiang, Feng, Linghao, Dong, Yiting, Wang, Jihang, Zhang, Qian, and Zeng, Yi. "Neuro-Vision to Language: Image Reconstruction and Interaction via Non-invasive Brain Recordings." *Proceedings of the 38th Conference on Neural Information Processing Systems (NeurIPS 2024)*, 2024. `Multimodal LLM` `fMRI` `NeurIPS`

5. **Shen, Guobin**, Zhao, Dongcheng, Bao, Aorigele, He, Xiang, Dong, Yiting, and Zeng, Yi. "StressPrompt: Does Stress Impact Large Language Models and Human Performance Similarly?" *Proceedings of the 39th AAAI Conference on Artificial Intelligence (AAAI 2025)*, 2025. `LLM Analysis` `Cognitive Science` `AAAI`

6. **Shen, Guobin**, Zhao, Dongcheng, Dong, Yiting, Zhang, Qian, and Zeng, Yi. "Alignment between Brains and AI: Evidence for Convergent Evolution across Modalities, Scales and Training Trajectories." *arXiv preprint arXiv:2507.01966*, 2025. `Brain-AI Alignment` `Cognitive Science` `Under Review in Nat. Hum. Behav.`

7. **Shen, Guobin**, Zhao, Dongcheng, Dong, Yiting, Li, Yang, Li, Jindong, Sun, Kang, and Zeng, Yi. "Astrocyte-Enabled Advancements in Spiking Neural Networks for Large Language Modeling." *arXiv preprint arXiv:2312.07625*, 2023. `LLM Pre-training`

Neural Network Accelerators

8. Li, Jindong, Li, Tenglong, **Shen, Guobin**, Zhao, Dongcheng, Zhang, Qian, and Zeng, Yi. "Hummingbird+: Advancing FPGA-based LLM Deployment from Research Prototype to Edge Product." *Proceedings of the 2026 ACM/SIGDA International Symposium on Field Programmable Gate Arrays (FPGA)*, 2026, pp. 235-246. `LLM Acceleration` `Edge AI` `FPGA`

9. Li, Jindong, Li, Tenglong, Chen, Ruiqi, **Shen, Guobin**, Zhao, Dongcheng, Zhang, Qian, and Zeng, Yi. "Hummingbird: A Smaller and Faster Large Language Model Accelerator on Embedded FPGA." *Proceedings of the 2025 International Conference on Computer-Aided Design (ICCAD)*, 2025. `LLM Acceleration` `FPGA` `ICCAD`

10. **Shen, Guobin**, Li, Jindong, Li, Tenglong, Zhao, Dongcheng, and Zeng, Yi. "SpikePack: Enhanced Information Flow in Spiking Neural Networks with High Hardware Compatibility." *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025. `SNN` `Hardware` `ICCV`

11. Li, Jindong, **Shen, Guobin**, Zhao, Dongcheng, Zhang, Qian, and Zeng, Yi. "Firefly: A High-Throughput Hardware Accelerator for Spiking Neural Networks with Efficient DSP and Memory Optimization." *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, vol. 31, no. 8, 2023, pp. 1178-1191. `Hardware` `Accelerator` `TVLSI`

Brain-Inspired AI & Spiking Neural Networks

12. **Shen, Guobin**, Zhao, Dongcheng, Dong, Yiting, and Zeng, Yi. "Brain-Inspired Neural Circuit Evolution for Spiking Neural Networks." *Proceedings of the National Academy of Sciences*, vol. 120, no. 39, 2023, p. e2218173120. `Neuro-Evolution` `SNN` `PNAS`

13. **Shen, Guobin**, Zhao, Dongcheng, and Zeng, Yi. "Backpropagation with Biologically Plausible Spatiotemporal Adjustment for Training Deep Spiking Neural Networks." *Cell Patterns*, vol. 3, no. 6, 2022. `SNN` `Backpropagation` `Cell Patterns`

代表性论文 (continued)

14. **Shen, Guobin**, Zhao, Dongcheng, Li, Tenglong, Li, Jindong, and Zeng, Yi. "Are Conventional SNNs Really Efficient? A Perspective from Network Quantization." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 27538-27547. `SNN` `Efficiency` `CVPR Highlight`

15. **Shen, Guobin**, Zhao, Dongcheng, Dong, Yiting, Li, Yang, Zhao, Feifei, and Zeng, Yi. "Learning the Plasticity: Plasticity-Driven Learning Framework in Spiking Neural Networks." *Proceedings of the 39th Conference on Neural Information Processing Systems (NeurIPS 2025)*, 2025. `Plasticity` `Learning Framework` `NeurIPS`

16. **Shen, Guobin**, Zhao, Dongcheng, and Zeng, Yi. "Exploiting High-Performance Spiking Neural Networks with Efficient Spiking Patterns." *IEEE Transactions on Emerging Topics in Computational Intelligence*, 2025. `SNN` `Efficiency` `TETCI`

Open-Source Frameworks

17. **Shen, Guobin**, Zhao, Dongcheng, Feng, Linghao, He, Xiang, Wang, Jihang, Shen, Sicheng, Tong, Haibo, Dong, Yiting, Li, Jindong, Zheng, Xiang, and others. "PandaGuard: Systematic Evaluation of LLM Safety in the Era of Jailbreaking Attacks." *arXiv preprint arXiv:2505.13862*, 2025. `AI Safety` `Framework`

18. Zeng, Yi, Zhao, Dongcheng, Zhao, Feifei, **Shen, Guobin**, Dong, Yiting, Lu, Enmeng, Zhang, Qian, Sun, Yinqian, Liang, Qian, Zhao, Yuxuan, and others. "BrainCog: A Spiking Neural Network Based, Brain-Inspired Cognitive Intelligence Engine for Brain-Inspired AI and Brain Simulation." *Cell Patterns*, vol. 4, no. 8, 2023, p. 100789. `Framework` `Brain-inspired` `Cell Patterns`

项目经历

- **PandaGuard: 大语言模型安全评估框架** Stars 66 **项目负责人**
  设计并实现系统性 LLM 越狱攻击安全评估框架, 集成多种攻击和防御算法
  构建大规模基准数据集 PandaBench, 提供多维度安全评估指标体系
- **BrainCog: 脑启发认知智能引擎** Stars 615 **项目负责人**
  领导开发综合性脉冲神经网络框架, 支持脑启发 AI 和脑仿真研究
  实现 50+ 功能性 SNN 算法, 涵盖感知学习、决策制定、知识表示等认知功能
- **aw_nas: 模块化神经架构搜索框架** Stars 252 **算法贡献者**
  为开源 NAS 框架贡献 Once-for-All 等算法的复现, 以及相关数据集的构建.

奖励与荣誉

2025.06 **中科院院长奖学金** (~1%) 非应届毕业生唯一获奖

2024.11 **博士生国家奖学金** (~1%)

2023.11 **Cell Press 年度论文** (学生一作)

2022.11 **Cell Press 中国科学家最佳论文奖** (第一作者)